Film Investment Screener: A Multimodal Machine-Learning Tool for Box-Office Prediction and Investment Decision Support

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Abstract

This capstone project develops a machine-learning decision-support tool for film investment, building on my previous work predicting box-office revenue. It investigates the effectiveness of multimodal film data, such as plot text, genre, production details, director track records, release timing, and early audience signals, in predicting box-office performance to assist with investment screening. Research indicates that combining features related to “who,” “what,” and “when” yields better predictions (Lash & Zhao, 2016). The project includes a supervised learning pipeline to predict logarithmic box-office revenue and a profitability classifier to assess whether a film will earn over 2.5 times its budget. An interactive Gradio web interface, deployed on Hugging Face Spaces, allows users to input a film concept and receive predicted revenue ranges, profitability probabilities, comparable films, and risk flags. While the model proves useful as an investment screener, accurately predicting revenue remains challenging for certain film categories.

Introduction

Investing in films is inherently uncertain because a film’s commercial performance relies on a mix of creative, industrial, and market factors. Investors are generally not focused on predicting the exact box-office numbers. Instead, they need to understand whether a project is likely to meet certain financial thresholds, if it falls within a commercially viable genre or release window, and how its risk profile compares to those of past successful or unsuccessful films. My capstone project addresses this challenge by developing a machine-learning system that translates film attributes into predictions relevant to investors. For me, the central problem was not simply whether a model could predict box-office revenue, but whether the prediction could be translated into a decision that a potential investor could actually use.

The research question guiding this project is: Can a multimodal machine-learning model that incorporates a film's plot, genre, cast, and director metadata, along with early audience signals, accurately predict box-office performance to create actionable investment decision rules for potential film investors? This question arises directly from my proposal, which positions the project within the field of media economics and frames box-office forecasting as both a supervised prediction problem and an investment decision problem.

This project builds on previous work in box-office prediction. Lash and Zhao (2016) argue that predicting early movie investments benefits from combining multiple feature groups: the individuals involved in the film, the film's content, and the release timing. Qiu and Zheng (2023) demonstrate that combining sentiment-driven forecasts can enhance box-office projections, particularly when model uncertainty is addressed carefully. Similarly, Cheng and Yang (2022) find that online review sentiment and review volume can improve economic models of box-office sales. Collectively, these studies suggest that predicting film performance should not rely solely on genre or budget; rather, it should integrate text, structured metadata, audience responses, and financial variables.

Data and Feature Construction

The project utilizes a comprehensive dataset that combines textual, categorical, and numerical features related to movies. Initially, several potential data sources were identified, including MovieLens 25M, TMDb metadata, IMDb review data, and box-office revenue datasets. These sources were selected for their diverse information relevant to film performance, such as audience behavior, genre, plot descriptions, cast and crew details, budget, revenue, and review or popularity indicators.

In the final implementation, I focused on a cleaned dataset that included film titles, release years, budget, total box office, genres, keywords, overviews, taglines, director information, original language, release months, and variables related to popularity or ratings. Textual features were created by combining genres, keywords, overviews, and taglines into a single text field, which was then vectorized using TF-IDF. Numerical features included the logarithm of the budget, runtime, logarithm of the vote count, vote average, popularity, average previous revenue of directors (logged), the number of previous films directed, release month, and indicators for summer or holiday release windows. Categorical features comprised the primary genre, original language, and release quarter. The final feature matrix integrated text, numerical, and categorical information into a cohesive input for the model.

This multimodal approach directly addresses a limitation from my earlier course project. In that earlier version, I primarily utilized genres and keywords, while some numerical variables were prepared but not fully incorporated into the final analysis. The capstone version of the project expands the model to systematically include budget, runtime, release timing, director history, and audience signals. This shift aligns the model more closely with real investment considerations, since revenue alone is less meaningful without accounting for budget and expected returns.

Methods

I approached the project as two interconnected supervised learning tasks. The first task was regression, which involved predicting the log-transformed box-office revenue. The log transformation was necessary because film revenue is highly skewed, with a few blockbuster films earning significantly more than most other releases. The second task was classification, where I aimed to predict whether a film is likely to be profitable, defined as earning more than 2.5 times its production budget. This profitability threshold serves as a simplified rule of thumb, making the output easier to interpret for investment screening.

In the modeling pipeline, I compared text-only baselines with fused multimodal models. The text-only baseline used TF-IDF features with a linear model. In contrast, the fused models combined TF-IDF text features with scaled numerical features and one-hot-encoded categorical features. The strongest model employed a gradient boosting regressor on the fused feature set. Additionally, I trained a logistic regression classifier to determine profitability for films with known budgets. Beyond standard regression metrics, such as R^2 and error measures, I also evaluated the model using investor-oriented metrics, including top-decile precision, decile lift, and simulated return on investment.

A significant methodological change from the earlier course project was the implementation of time-aware evaluation. Rather than relying on a random train/test split, which can overestimate performance by mixing older and newer market conditions, I opted for a chronological split. This approach better reflects the real-world scenario where investors train on past films and make decisions about future releases. This change was important because my earlier random-split model treated the dataset as if all films came from the same market environment, while a real investor would always be making decisions about future releases based on past films. The proposal specifically identified time-based leakage and temporal-domain shift as critical issues in film investment prediction, and the final project was designed to address these concerns.

Results

The results indicate that the model is more effective as a screening and ranking tool than as an accurate revenue forecaster. In time-aware testing, the text-only TF-IDF baseline performed poorly, resulting in a negative test R^2. This suggests that the strong performance observed in the earlier course project may have been inflated by leakage or by the similarity between randomly mixed films. When the model was required to generalize to newer films, text alone proved insufficient. This result became one of the main reasons I shifted the project away from exact revenue prediction and toward screening, ranking, and risk interpretation.

In contrast, the fused multimodal model showed significant improvement. The best-performing model was the gradient boosting model that utilized a combination of text, numerical, and categorical features. In the final notebook summary, this model achieved a test R^2 of approximately 0.33 on log revenue, with a precision of 73% among the top 10% of predicted films, a decile lift of about 5.9×, and a simulated ROI of 3.57× on a top 25 portfolio. These results suggest that, even though the model may not perfectly predict exact revenue, it can effectively rank films for portfolio-style investment screening.

The profitability classifier performed particularly well, achieving a validation AUC of approximately 0.80 and a test AUC of around 0.88. Since this classifier predicts whether a film surpasses the 2.5× budget threshold, its output is more actionable than raw revenue predictions. A film with a high predicted profitability is a strong candidate for greenlighting. In contrast, a film with a low probability can be flagged for rejection or require a more in-depth qualitative review.

Finally, the unsupervised analysis revealed interpretable genre-level patterns. K-means clustering and LDA topic modeling indicated that family animation, fantasy adventure, action, and science-fiction content tend to occupy more commercially promising areas of the dataset. In contrast, romance and smaller-scale drama clusters show lower average revenue. These findings do not imply that genre causes revenue, but they do demonstrate that content patterns derived from text features correspond to meaningful commercial differences.

Interactive User Interface

A key deliverable of this capstone project is the Film Investment Screener, which features an interactive Gradio interface. This user-friendly UI enables users to input various details about a hypothetical film, including the title, keywords, budget, release month, primary genre, logline, tagline, runtime, original language, director's track record, prior films, vote count, vote average, and popularity. Once the user submits the information, the app generates four outputs: predicted revenue along with an 80% confidence interval, the probability of profitability, comparable historical films, and risk flags. The interface utilizes a trained gradient boosting model, a profitability classifier, feature pipelines, residuals, and a reference film dataset sourced from saved artifacts.

The application is designed to be accessible for non-technical users. Rather than presenting only numerical predictions, it offers a comprehensive screening report featuring visual cards. This includes a green/yellow/red profitability interpretation, a break-even target, and comparable films identified through TF-IDF cosine similarity. The risk flags are based on known model failure modes, such as missing budget information, extremely low or high budgets, non-English films, less favorable release months, high-budget drama or romance projects, unknown directors, and unusually wide confidence intervals. The final user interface was uploaded to both GitHub and Hugging Face Spaces, which can be accessed at: https://huggingface.co/spaces/joezhou21/film-investment-screener.

Discussion

The main finding of this capstone project is that multimodal machine learning can support film investment decisions, but only when the task is framed realistically. In this sense, the project shifted focus from finding a perfect forecasting model to developing a structured approach for distinguishing more promising projects from riskier ones. Predicting box-office performance with precision remains challenging because of the myriad factors that influence film success, many of which are not fully captured in public datasets. These factors include marketing expenditure, distribution strategy, star publicity, franchise expectations, platform competition, and cultural timing. However, investors do not necessarily require a perfect point prediction. Instead, they need a model that can rank projects, flag risks, identify comparable films, and estimate whether a film has a reasonable chance of becoming profitable.

This project argues that incorporating multimodal features is essential for meaningful predictions. Text features help the model understand the film's content, while budget and release metadata provide critical insights into scale, distribution goals, and market positioning. Additionally, information about the director's history and audience signals contributes another layer of understanding regarding the expected commercial potential. This aligns with Lash and Zhao's (2016) focus on the importance of "who," "what," and "when" features, as well as the value of sentiment and review-based variables highlighted in more recent research (Cheng & Yang, 2022; Qiu & Zheng, 2023).

Furthermore, this project underscores the importance of time-aware evaluation. A random split of data can create an illusion of model strength, as films from similar periods and market conditions may appear in both the training and test sets. The film market post-2020 is markedly different due to the rise of streaming releases, pandemic disruptions, changes in theatrical attendance, and varying distribution strategies. By utilizing a chronological split, this capstone provides a more accurate estimate of how the model would perform in a real investment scenario.

Implications

This project has practical implications for the use of machine learning in film development and investment decisions. The Film Investment Screener serves as an early-stage tool that helps compare hypothetical film concepts, identify commercially promising patterns, and flag projects that may require closer examination. For example, it can help users evaluate whether a film's genre, budget, release timing, and textual profile resemble those of past high-performing films. It also provides warnings when predictions may be less reliable due to missing budget data, non-English-language films, or unusually wide confidence intervals.

Additionally, the project suggests that film investment should be assessed from both predictive and interpretive perspectives. Even if a model cannot perfectly predict exact revenue, it can still help formulate better investment questions, such as: What comparable films does this project resemble? Is the budget appropriate for the genre? Does the release window increase or decrease risk? Is the predicted profitability strong enough to justify further development? In this way, the screener's value lies not only in providing forecasts but also in organizing the decision-making process around comparisons of risk and expected return.

More broadly, this capstone demonstrates how academic prediction models can be transformed into accessible tools for creative industries. By deploying the model through a Gradio interface on Hugging Face Spaces, the project converts a technical notebook into a user-friendly platform that non-technical users can test and interpret. This approach is significant because cultural products are influenced by timing, taste, marketing, and human judgment in ways that public data alone cannot fully capture. Thus, machine learning may be most valuable in creative markets when it supports exploration and risk awareness rather than asserting final decisions.

Limitations

Several limitations are present in the model. First, it does not account for marketing spend, which is one of the most significant predictors of box-office performance, yet such data is rarely available in public datasets. Second, budget information is often incomplete, particularly for independent and international films. Since profitability classification relies on the budget, the classifier performs best for films with more comprehensive production data. Third, the model may overpredict the success of re-releases or films with a substantial historical popularity, as the popularity metrics may reflect the original release rather than the specific theatrical run being evaluated. Fourth, the model's reliability decreases for non-English films and limited-distribution releases, where global popularity metrics may not directly correlate with U.S. or worldwide theatrical revenue. Lastly, it is important to note that the model is correlational rather than causal. A high-budget film might generate more revenue not solely because of its budget, but also because it is linked to studio support, marketing, distribution, talent, and franchise value. Consequently, the model should be utilized as a decision-support tool rather than as a definitive greenlight system.

References

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